Day 6 : Introduction to Network Devices Theory

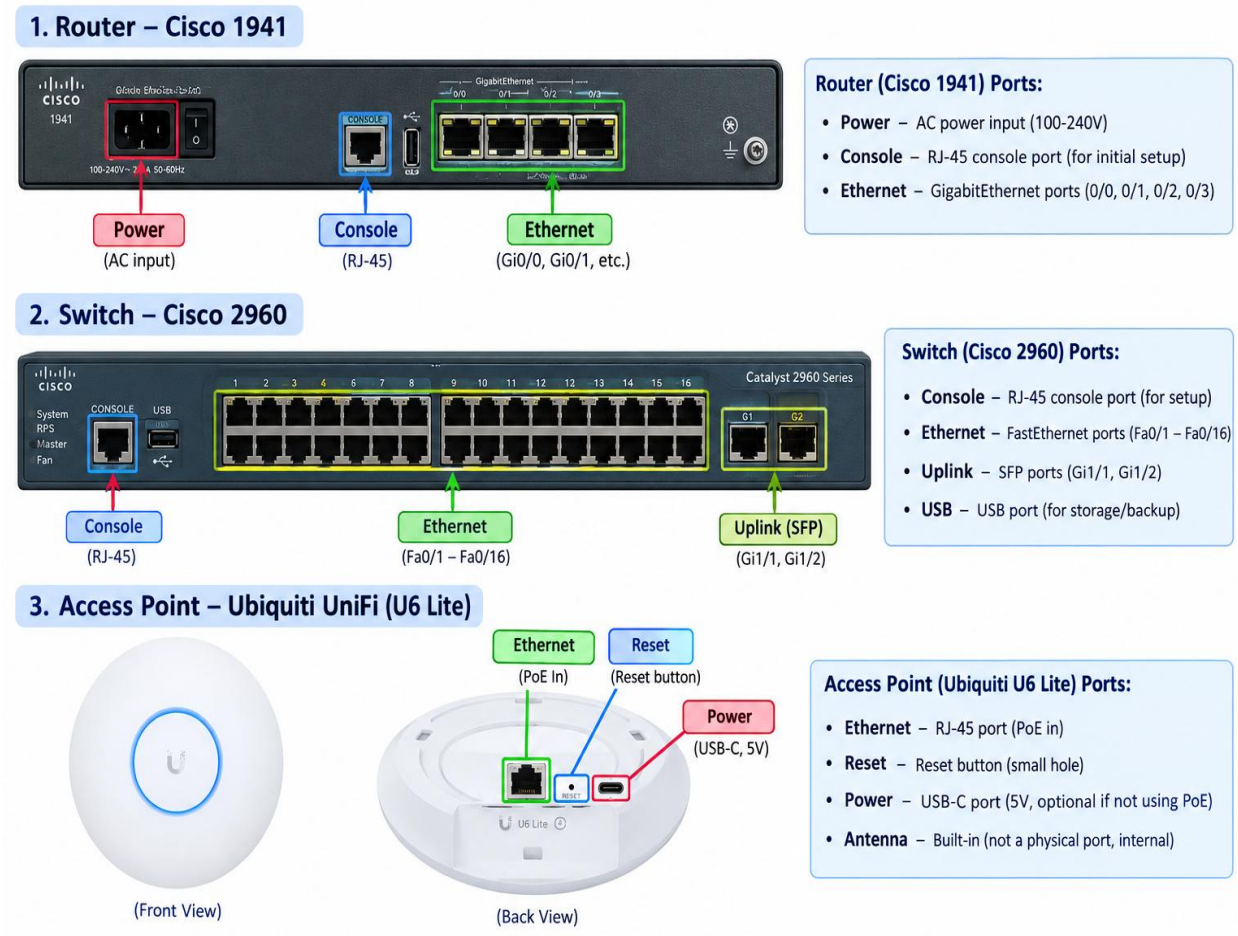
Tasks 1 : Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

Network Device	Main Function	Typical Ports / Interfaces	Use in a Home Network
Switch	Connects multiple wired devices within the same local area network (LAN) and forwards data to the appropriate port.	Multiple Ethernet ports (RJ-45); some switches also have uplink or SFP ports.	Connects desktop computers, printers, smart TVs, and other wired devices.
Router	Connects different networks and forwards packets between them. It commonly provides NAT, DHCP, and basic firewall features in home networks.	WAN Ethernet port, LAN Ethernet ports, and sometimes a console or USB port.	Connects the home network to the internet through the ISP.
Hub	Sends incoming data to all other ports instead of forwarding it only to the intended device.	Multiple Ethernet ports (RJ-45); some older models have a power connector.	An older way to connect wired devices; largely replaced by switches.
Firewall	Monitors and filters network traffic according to security rules.	Ethernet interfaces; some dedicated models also include console, management, or USB ports.	Helps block unauthorized connections and protect devices from unwanted network traffic.
Wireless Access Point (AP)	Allows wireless devices to connect to a wired LAN using Wi-Fi.	Ethernet port (often supporting PoE), power input if required, and internal or external antennas.	Provides Wi-Fi coverage for phones, laptops, tablets, and smart devices.

Important Notes

1. A **WAN port** on a router typically connects to the ISP modem or internet gateway, while **LAN ports** connect to the local network.
2. A **console port** is generally used for device configuration and management, not for normal user data traffic.
3. An access point may receive power through a separate adapter or Power over Ethernet (PoE).
4. Antennas are used for wireless transmission and reception; they are not network ports.
5. Many home Wi-Fi routers combine router, switch, access point, and basic firewall functions in a single device.

Task 2 : Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.



1. Router

Device: Router

- **WAN Ethernet Port:** Connects the router to the modem or ISP gateway.
- **LAN Ethernet Ports:** Connect wired devices such as computers and switches.
- **Power Port:** Supplies electrical power to the router.
- **Antennas:** Transmit and receive Wi-Fi signals, if the router has external antennas.
- **Reset Button:** Restores factory settings when used according to the manufacturer's instructions.

2. Network Switch

Device: Ethernet Switch

- **Ethernet Ports (RJ-45):** Connect computers, printers, routers, and other wired network devices.
- **Uplink Port:** Connects the switch to another switch or a router. On many modern switches, any suitable Ethernet port can serve this purpose.
- **Power Port:** Supplies power to the switch.
- **Console Port:** Available on some managed switches for configuration and troubleshooting.

3. Wireless Access Point

Device: Wireless Access Point (AP)

- **Ethernet Port:** Connects the access point to the wired network.
- **PoE Port:** On supported models, receives both data and electrical power through one Ethernet cable.
- **Power Port:** Supplies power through an adapter when PoE is not used.
- **Antennas:** Transmit and receive Wi-Fi signals; they may be internal or external.
- **Reset Button:** Restores factory settings when required.

Task 3 : Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

Application: Zomato

- Zomato uses several network devices to deliver food-ordering services to users.
- A **wireless access point (AP)** connects a user's mobile phone to the Wi-Fi network.
- The **router** forwards the user's request to the internet and routes the response back to the device.
- **Switches** connect multiple devices within a network, while **firewalls** help protect servers and networks by filtering unauthorized traffic.
- These devices work together to make Zomato accessible and secure.

Task 4 : Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Problem: College Wi-Fi is slow and frequently disconnects.

Possible Reason:
One possible reason is that too many students are connected to a limited number of old or low-capacity wireless access points. This can cause network congestion, slow internet speeds, and frequent disconnections.

How Upgrading Network Devices Helps:

1. **Wireless Access Point (AP):** Upgrading to modern Wi-Fi 6 access points can support more connected devices and improve network performance.
2. **Switch:** Upgrading to a high-capacity Gigabit or multi-gigabit switch can provide faster connections between access points and the main network.
3. **Router:** Upgrading the router can improve traffic handling and prevent it from becoming a bottleneck when many students use the internet simultaneously.
4. **AP Placement:** Installing additional access points in crowded areas can improve signal strength and reduce congestion.

Conclusion:
Upgrading outdated network devices, improving access point placement, and providing sufficient network capacity can make college Wi-Fi faster, more reliable, and capable of supporting many students simultaneously.